

# SCA: Leveraging Google Cloud to democratize data in a secure, highly available in reliable fashion for our entire business

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ABOUT SOUTHERN CROSS



About Southern Cross  
Austereo

SCA optimizes developer productivity and costs by consolidating developer and data on Google Cloud, VMs on Compute

SCA is Australia's leading media company, reaching more than 95% of the national population through its radio, television and digital assets. The radio and audio network launched LiSTNR as a free, personalized audio destination for all its digital audio assets, including Triple M and Hit stations. The LiSTNR app gives people the freedom to listen to their favorite programs on any Android or iOS device at any time, anywhere. SCA fosters a culture of innovation and sustainability through its in-house

power data processing  
pipelines providing  
quality business results

innovation program, The Lab, where employees  
can submit creative ideas for LiSTNR.

Industries: Media & Entertainment

Location: Australia

## Google Cloud results

- Realise value from data by building innovative and efficient data led capabilities
- Promotes new use cases with rich data sets with Data Catalog
- Delivers high availability across multiple cloud regions in Australia

## Optimised with pe billing pipelin

(<https://cloud.google.com/composer/>), [Cloud](#)

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/functions/>),

BigQuery (<https://cloud.google.com/bigquery/>) and

IAM (<https://cloud.google.com/iam/>), Data

Cloud Composer

(<https://cloud.google.com/composer/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

(<https://cloud.google.com/composer/>),

Firebase (<https://firebase.google.com/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Data Catalog

(<https://cloud.google.com/data-catalog/>)

Contact us ([/contact](#))

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

Compute Engine

(<https://cloud.google.com/compute/>)

**for scaling  
the  
business**

Traditionally, live radio follows a fixed schedule, so if a listener tunes in halfway through a program, they miss the show's first half. Thanks to digital audio apps, it is now possible for people to play many of their favorite shows whenever they want, and where they want, such as on an Android Auto player in the car or a smart speaker at home.

Addressing digital consumption, [Southern Cross Austereo \(SCA\)](https://www.southerncrossaustereo.com.au/).

(<https://www.southerncrossaustereo.com.au/>), a commercial broadcasting company, launched the LiSTNR digital platform so consumers can log in and listen to radio shows, podcasts, music, and news, live or on-demand. Powered by [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>), LiSTNR provides SCA with real-time insights to respond quickly to changes in listening habits and market demands.

"LiSTNR is key to SCA's digital transformation strategy as a fully digital and measurable platform," says Leon Kelly, Head of Digital Technology at Southern Cross Austereo. "Google Cloud runs our data pipelines at scale, so we tailor content based on listeners' routines, needs, and preferences. Our vision is to create a personalized listening experience that entertains, informs, and inspires Australians."

Business continuity is vital for SCA as audience size and time spent grows month by month. The company gains high availability within one cloud provider by migrating LiSTNR from an on-premise data center to a fully managed data stack across two Google Cloud regions in Melbourne and Sydney.

According to Kelly, "Part of my responsibility is to identify and eliminate single points of failure. Having redundancy built in our data architecture on Google Cloud helps me sleep at night, knowing that we can keep operations running smoothly no matter what happens."

SCA used Airflow on Cloud Composer (<https://cloud.google.com/composer>). As a fully managed orchestration tool, Cloud Composer increases SCA's IT agility, allowing developers to quickly move data pipelines from one environment to another based on their requirements, and automate workflows.

SCA's data architecture in Google Cloud also includes Dataproc (<https://cloud.google.com/dataproc>) for big data processing, Cloud Storage (<https://cloud.google.com/storage>) for storing data, and BigQuery (<https://cloud.google.com/architecture/bigquery-data-warehouse>) as the data warehouse to power business intelligence dashboards.

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Digital Technology,

## **Improving the listener experience with personalized content**

Just as e-commerce platforms offer product recommendations to help customers find relevant and targeted products, LiSTNR delivers personalized content when users log on. For example, SCA shares local news specific to the listener's postcode and recommends new content based on their preferences.

In the past, processing large datasets for recommendations and personalization algorithms was not cost effective. SCA paid by the hour whenever it moved data from the legacy data platform into Cloud Storage and BigQuery for analysis.

"From the beginning, we leveraged Google Cloud for our DevStack, DataProc to leverage its high quality data processing and keeping data in cost effective, highly available Google Cloud Storage. To optimize costs, it made sense to bring the DevStack and data stack on the same cloud framework to enjoy 'per minute' basis," says Abhishek Bandyopadhyay, Data Engineering Manager at Southern Cross Austereo.

"From a manageability standpoint, we're less dependent on external teams to maintain the data on Google Cloud so we can focus on customer requirements."

SCA also brings up 30+ workflows daily in Dataproc for high quality, reliable data processing without worrying about setting up the environment. SCA can quickly stop instances when they are not needed so that it can save on cloud costs.



## **Optimizing ad performance with targeted custom personas**

According to a GFK report, "The number of Australians listening to digital audio has doubled since 2016 and is expected to reach 80% of Australians by 2024." Digital audio advertising presents a growing opportunity for brands to engage with existing and new customers.

Radio advertising is traditionally a push marketing tactic, where advertisers promote their brand to a mass audience with limited ability to track results. Ad companies relied on third-party cookies to track online customers across platforms. As popular internet browsers gradually phase out third-party cookies, LiSTNR must leverage first-party data to

help advertisers identify and target specific audiences for their digital audio ads.

When people use LiSTNR, SCA saves their activity, including listening time and device type as data, with opt-in consent. Using privacy-compliant data, SCA creates profiles of customers who share similar attributes so brands can match ads to their target audience for optimal results.

"Moving to a first-party world to know more about our customers has been our number one priority for the past six months. Having a rich customer data set on BigQuery helps us understand how customers interact with us so we can provide a great user experience through personalized content and relevant ads," says Kelly.

Apart from real-time app analytics in Firebase, SCA integrates more than 35 customer touchpoints, from contact forms to radio call-in contests, for a detailed picture of the customer journey. SCA integrates first-party data with third-party sources such as Google Analytics for mobile app data and HomeStudio for podcast log information to track and measure how listeners react to audio ad campaigns.

"From the  
beginning, we  
leveraged



Google Cloud for  
our DevStack,  
DataProc to  
leverage its high  
quality data  
processing and  
keeping data in  
cost effective,  
highly available  
Google Cloud  
Storage. To  
optimize costs, it  
made sense to  
bring the  
DevStack and  
data stack on  
the same cloud  
framework to  
enjoy 'per  
minute' basis."

—Abhishek  
Bandyopadhyay, Data  
Engineering Manager,  
Southern Cross Austereo

**Driving business value with data  
discovery on Data Catalog**

For brands to create value from data, they first need to ingest the required data assets and make it available for everyone to be able to find it. In order to do so, SCA adopts Data Catalog (<https://cloud.google.com/data-catalog>) as a central place for teams to discover data assets that are available in its organization for creating new use cases, be it a customer feature or an internal process.

"Using Data Catalog, we make it easy for everyone, from business users to data analysts, to search, discover and securely access the data they need without IT intervention," says Sarita Charde, Data Analyst at Southern Cross Austereo. "By tagging BigQuery datasets and tables with business metadata, data owners help non-technical users make sense of technical data."

Charde adds that business metadata provides context to technical data such as "Where is the data sourced from? What can you do with it? Who's the custodian?"

The email marketing subscriber dataset is the first use case included in SCA's Data Catalog. Because the data is essentially a subscriber list, it is tagged as Personally Identifiable Information (PII) to alert business users who might need to access it. SCA protects sensitive data with predefined Cloud Identity and Access Monitoring (<https://cloud.google.com/iam>) (IAM) roles for data privacy. For example, authorized external parties can view but not make any changes to the data.

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## Securing data exchange to uncover new opportunities

The private sector is increasingly taking the lead from governments who share public-sector data such as geospatial and transportation data to improve collaboration. According to Gartner, data and analytics leaders who promoted data sharing were 1.7 times more effective

(<https://www.gartner.com/smarterwithgartner/data-sharing-is-a-business-necessity-to-accelerate-digital-business>)

in providing enterprise value. To support data reuse for new services, SCA is exploring Analytics Hub (<https://cloud.google.com/analytics-hub>) for its library of internal and external data assets, powered by BigQuery.

"We're sharing data with internal business users and external parties such as AI/ML partners to enrich data insights," says Bandyopadhyay. "By sharing our data through Analytics Hub in the future, we can control access with security perimeters and track who uses the data for auditing purposes."

According to Kelly, "We'll continue to drive increased returns on investment from data products and services moving forward. Google Cloud supports machine learning at scale so we can leverage data insights to improve content for listeners and campaign results for advertisers as we grow."

